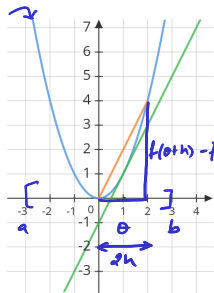


Derivative $f: \mathbb{R} \rightarrow \mathbb{R}$

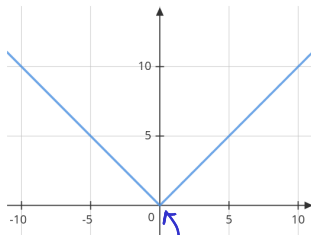
$f(\theta) = \theta^2$



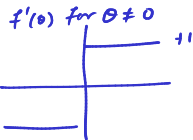
$\theta = 1$
 $h = 1$
 $\theta + h = 2$
 $\theta - h = 0$
 $f(\theta + h) = f(2) = 4$
 $f(\theta - h) = f(0) = 0$

Suppose $f(\theta)$ is defined on $[a, b]$. For $\theta \in [a, b]$, if the limit exists then we can define

$$\begin{aligned} \frac{df}{d\theta} &:= \lim_{h \rightarrow 0} \frac{f(\theta + h) - f(\theta - h)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{(\theta + h)^2 - (\theta - h)^2}{2h} = \frac{(1+1)^2 - (1-1)^2}{2} = \frac{4 - 0}{2} = 2 \\ &= \lim_{h \rightarrow 0} \frac{4h\theta}{2h} = 2\theta = f'(\theta) \end{aligned}$$

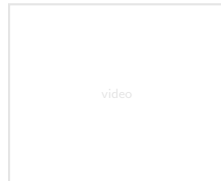


Derivative of the non-smooth function $f(\theta) = |\theta|$ does not exist at $\theta = 0$.

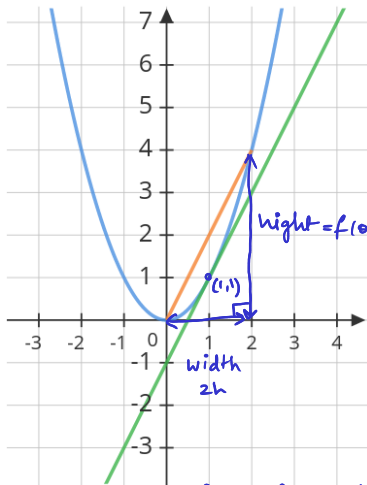


$$f'(\theta) = \begin{cases} \theta & \theta \geq 0 \\ -\theta & \theta < 0 \end{cases}$$

Step
 f''



Derivative



- Suppose $f(\theta)$ is defined on $[a, b]$.
For $\theta \in [a, b]$, if the limit exists then we can define

$$f'(\theta) = \frac{df}{d\theta} := \lim_{h \rightarrow 0} \frac{f(\theta+h) - f(\theta-h)}{2h}.$$

- $f'(\theta)$ is the **rate of change** of the function f with respect to the variable θ .

- Geometrically, $f'(\theta)$ is the **slope** of the **tangent line** to the function f at θ .

$$= \frac{\text{height}}{\text{width}} = \frac{4}{2} = 2$$

$$f(\theta) = \theta^2$$

$$f'(\theta) = 2\theta$$

Tangent line through

$(\theta_0, f) = (1, 1)$ is

$$f = 2\theta - 1$$

$f(\theta_0 + h) \approx f(\theta_0) + f'(\theta_0)h$
(truncated Taylor series)

$$\begin{aligned} f(\theta) &= f(1) + f'(1)(\theta - 1) \\ &= 1 + 2(\theta - 1) \end{aligned}$$

$$\theta_0 = 1$$

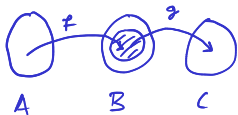
video

Q_{AI} Derivative Rules

1 **Product of functions:** $\frac{d(fg)}{d\theta} = f(\theta)g'(\theta) + f'(\theta)g(\theta)$

2 **Composite function** $h(\theta) = g(f(\theta))$ we have the chain rule $h'(\theta) = g'(f(\theta))f'(\theta)$

$$\phi = f(\theta)$$



function $f(\theta)$	derivative $f'(\theta)$
constant	0
Power θ^n	$n\theta^{n-1}$
exp e^θ	e^θ
$\ln(\theta)$	$\frac{1}{\theta}$
$\sin(\theta)$	$\cos(\theta)$
$\cos(\theta)$	$-\sin(\theta)$

$$= \frac{dg}{d\phi} \frac{d\phi}{d\theta}$$

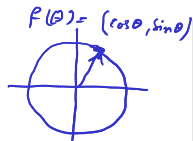
$\frac{1}{f}$

$\frac{1}{\theta}$

Vector valued f^n

For $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{R}^n$, where $\mathbf{f}(\theta) := (f_1(\theta), f_2(\theta), \dots, f_n(\theta))$, we have

$$\mathbf{f}'(\theta) = (f'_1(\theta), f'_2(\theta), \dots, f'_n(\theta))$$



video